

Geothermal surface solutions

Baker Hughes

Industrial Energy Technology

Our 40+ years of geothermal expertise ensures success

Baker Hughes has worked in geothermal for more than 40 years, delivering subsurface and surface expertise – and well as compliant operations –and helping you clearly understand and mitigate the risks.

Planning, designing and executing well construction projects in extreme environments

- Our subsurface experts, located in nearly every geothermal region, offer a rich mix of skill in hot reservoirs, geomechanics, and reservoir chemistry, among others.
- Our technical team can determine the perfect technology fit for each application.

Surface production, both reliably conveying the heat transfer and producing power from it

Our experience as an original equipment manufacturer in power generation, plant management, control systems and condition monitoring – combined with our understanding of the subsurface – allows us to properly match the power-generating capacity to the subsurface resource.

Efficient operations that comply with environmental, social, and governance requirements

Our commitment to ESG means that we will do the right thing, always – giving you confidence in how the project is executed.

Baker Hughes geothermal experience



Baker Hughes geothermal experience is global

USA: Enhanced production by 20 MW

- Characterization of in situ state of stress
- Geothermal production

Indonesia: Optimized well trajectories to increase well productivity

- Stress sensitive fractures
- Drilling and stimulation

USA: Exact wellbore place with minimal losses

- Deep directional drilling
- Basalt formation
- High temperature ~343°C

Finland: Saved 50 days of AFE

- Basement granite (UCS 85ksi)
- Deep directional drilling

USA: Characterized and qualified discovery well

- Stress field, natural fracturing, geomechanical properties
- Cross Multipole Array Acoustilog (XMAC™) service

Turkey: Saved 5 days of drilling

- High temperature ~250°C, 4509m deep
- Abrasive formation with high vibrations

Australia: Proved electricity generation feasible and profitable

- Wellbore instabilities
- Optimize production

Italy: Larderello Site

- 180 MW ST installed
- 16 compressors installed
- Drillbit applications

Iceland

- 300°C directional drilling system
- Deepest and hottest well ever drilled

New Zealand: Optimized plan for new well development

- High-temperature reservoir
- Wellbore instability and losses
- Local restrictions

Japan: Reduced AFE by 50%, saving \$2MM and enabling customer to drill one additional well

Igneous rocks, multiple bit trips, low ROP

Indonesia: Increased output up to 424 MW and saved \$25 million

- Geothermal production
- Wellbore instabilities

Germany: Saved 16 days of operation and 8 days of completion operations

- Deep directional drilling
- Precise wellbore placement

Indonesia: Saved 5 days of drilling

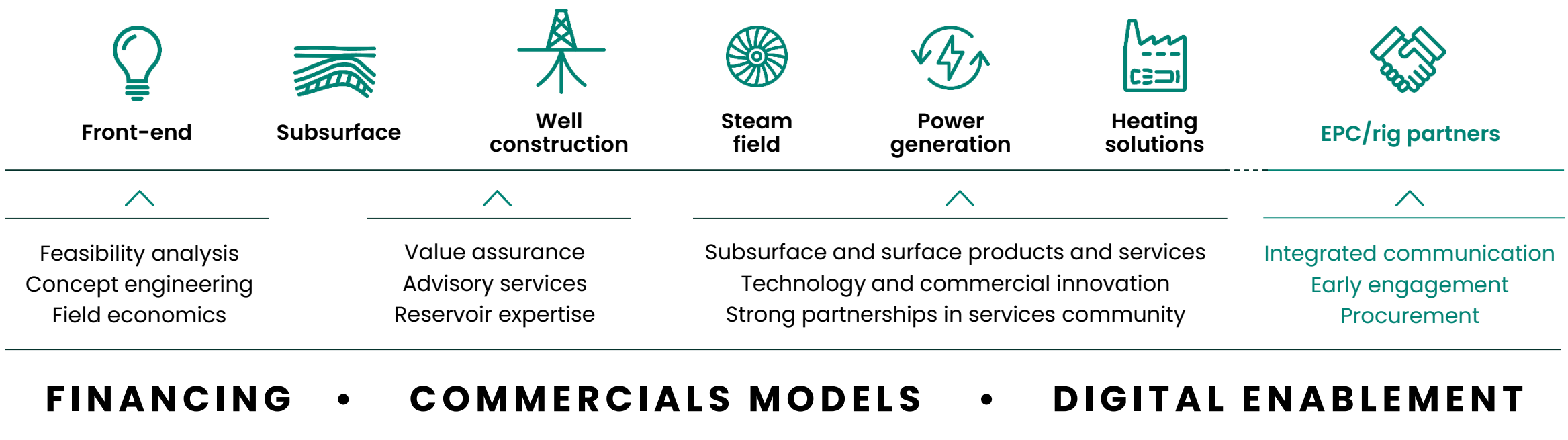
Igneous rocks, UCS 15–35kpsi, high temperature ~200°C

Philippines: Expanded power plant within existing development block

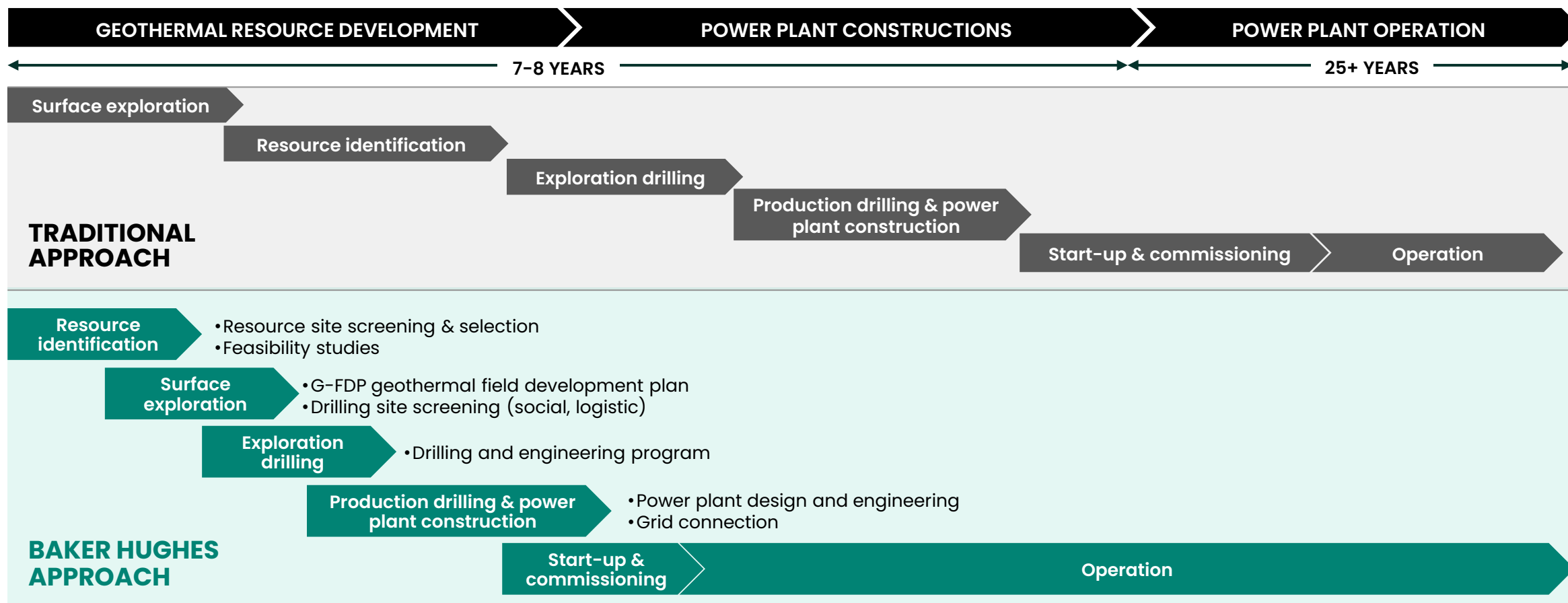
- Drilling and production
- Reservoir within existing permit area



Taking geothermal energy forward will require *integrating the subsurface with the surface*

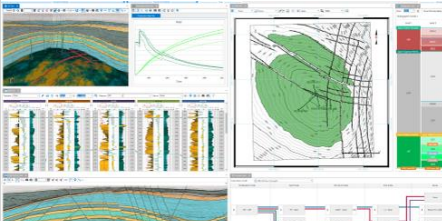
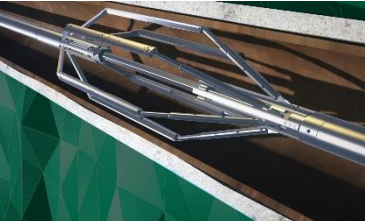


Our integrated approach mitigates risk and gets projects online faster



Geothermal Surface Solutions

Comprehensive subsurface-to-surface experience

DESIGN AND SOFTWARE SERVICES	SUBSURFACE SYSTEMS			SURFACE SYSTEMS	
					
RESERVOIR MODELING	WELL CONSTRUCTION	EVALUATION & MONITORING	COMPLETION & PRODUCTION	EQUIPMENT	PLANT ENGINEERING & MONITORING
JewelSuite™ subsurface modeling JewelSuite geomechanics JewelSuite reservoir modeling Connection to simulation engines	Drill bits Drilling services Drilling & completion fluids Cementing	Wireline services Coring Wellbore monitoring Integrated reservoir characterization	Completions & well intervention Hydraulic fracturing/stimulation Artificial lift Specialty chemicals	Surface trees Wellhead systems Flow control Field service	Steam turbines ORC / binary cycle Turboexpander generator Brush electric generators Brine pumps Valves Digital solutions NCG compressors Micro-seismic & fiber-optic monitoring

Baker Hughes – Steam Turbines

Le Creusot-France Thermodyn

Impulse Technology:

- Boiler feed Pump Drives
- Biomass
- Captive
- Navy

Range: 2 – 45 MW

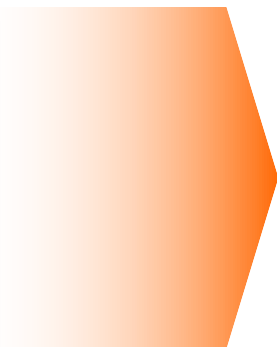


Firenze-Italia Nuovo Pignone

Reaction Technology:

- Compressors Drives
- Combined Cycles
- Biomass
- CSP
- **Geothermal**

Range: 2 – 150 MW



Model Steam parameters 0 20 40 60 80 100 120 140

SC/SAC/
SNC

140 bar
565°C



A5/A9

140 bar
540°C

Single casing reheat

SG/SDFG

30 bar
300°C

Geothermal ST

Geothermal STG – SG model

Plant Type	Geothermal / LP recovery
Power	• 5 to 80MW depending @ 50/60 Hz • Single & Double flow configuration • Condensing LP up to 6.8 m²
Technology	• Live steam up to 30 bar and 300 °C • Reaction type technology with control stage
Benefits	• Efficiency ... Outstanding heat rate due to material flexible and high performance steam path • effectiveness ... Low maintenance thanks to advanced moisture removal system and anti corrosion features • Simplicity ... Modular and compact concept, fabricated, welded structure with integral bearing pedestals



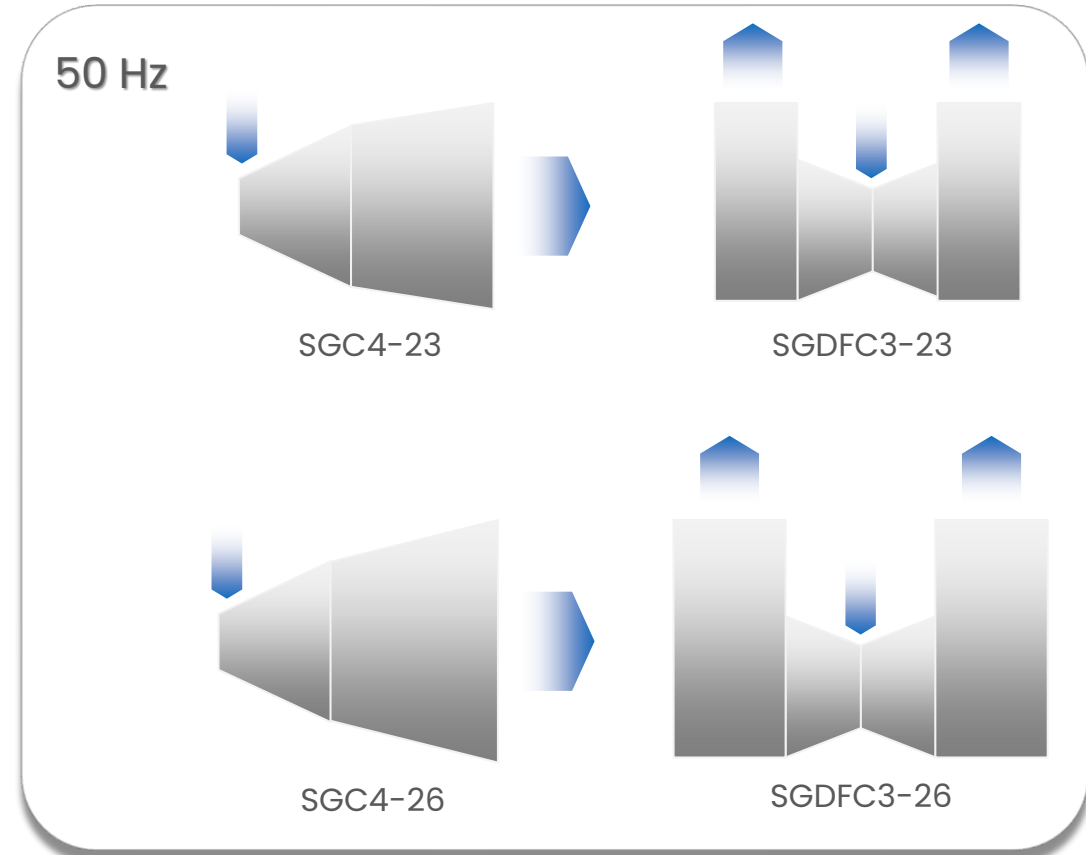
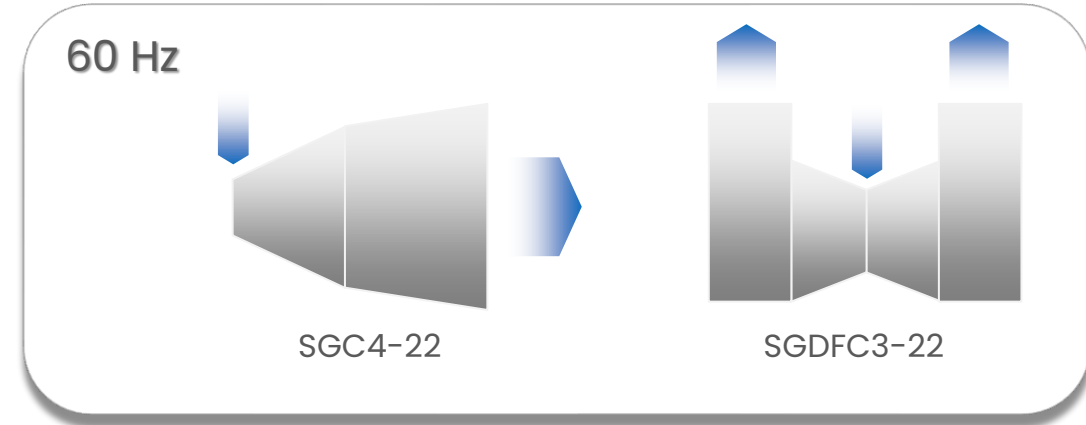
Steam Turbines for Geothermal

Equipment Scope – typical

- Steam Turbine and auxiliaries
- Generator
- Instrumentation and Unit Control Panel
- Condenser
- Erection and commissioning Technical Advisory or Supervision

Geothermal Turbine Highlights

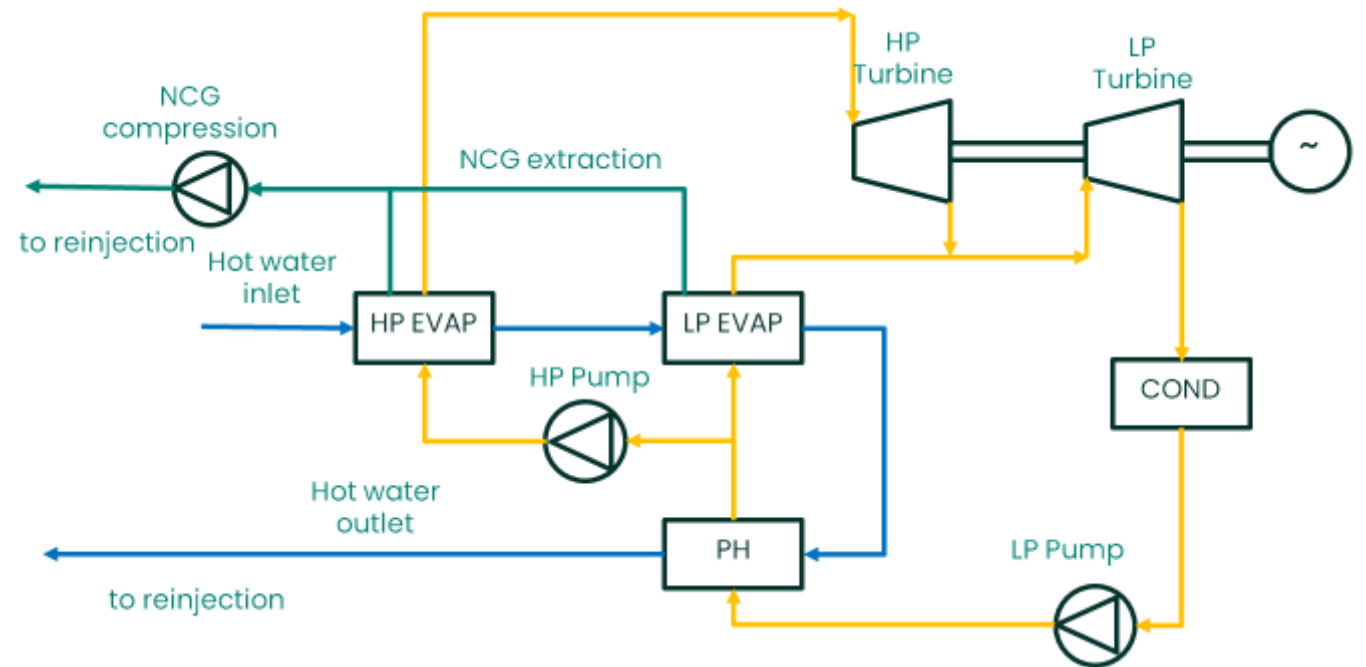
- Flexible architecture to cover a wide range of steam conditions and delivered power starting from predefined modules
- Customized reaction steam path to handle aggressive steam coming from geothermal fields
- Radial and axial exhaust
- Single and double flow
- 50 and 60 Hz



Medium Enthalpy ORC for geothermal application

Key drivers for ORC Application

- Geofluid temperature between 120°C and 210°C, liquid dominated
- Single or multi-pressure ORC for best efficiency
- Zero leakage solution with NCG reinjection
- Negligible OPEX:
 - Unmanned operation
 - Low maintenance



In-house Core Technology: Turboexpander-Generator, Air Cooled Condenser, Control Systems, Working Fluid Pumps

5-40 MW ORC solutions for medium enthalpy geothermal applications

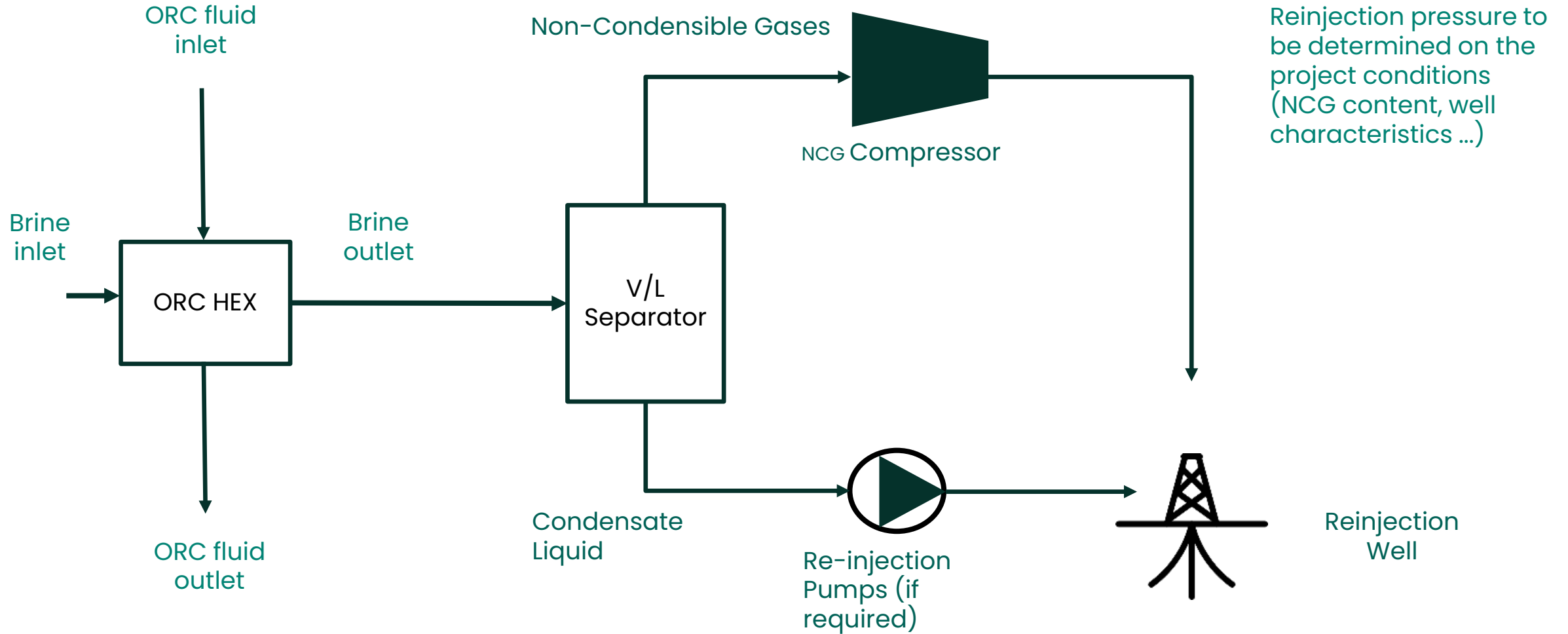
Turbo-Expander for waste heat recovery (ORC)

Two stages expander generator designed for ORC plant based on gas turbine exhaust heat recovery

Location	Whitecourt, Canada	Berakas, Brunei	Saiyok, Thailand
Gas	CC5	CC5	CC5
P in (barA)	39.1	39.3	39.2
T in (°C)	250	250	239
P out (barA)	1.3	1.4	1.5
T out (°C)	140	146	130
TOT EXP. Power (MW)	16.1	17.3	12.7
HP/LP Speed (rpm)	10180 / 4950	10188 / 4990	10180 / 4950
Shipped year	2012	2015	2015



NCG reinjection

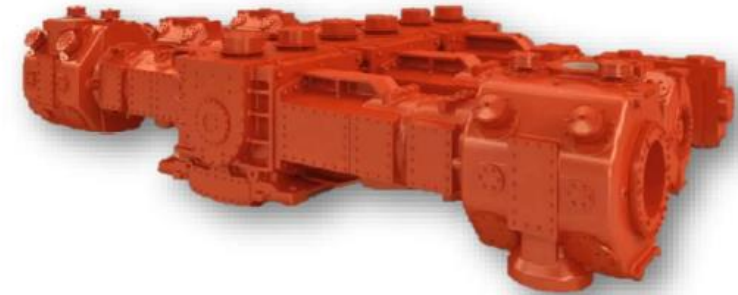


NCG reinjection and extraction compressor

Key features

- Reciprocating compressors for total NCG reinjection
 - Suitable to compress all NCG from CO₂ to H₂S
 - Capable to reach required reinjection pressure
 - Packaged solution
 - Up to 6 cylinder per unit
-
- Integrally geared compressors for NCG extraction from ST condenser
 - Capability to work under vacuum (minimum inlet pressure 0.06 barA)
 - Up to 2 MW power and 4 impellers

*NCG: "Non-Condensable Gases"



Pumps offering for Geothermal plants

	Boiler feed	Brine re-injection	Cooling water	Boosting	Condensate	Hot Fluid transfer	District heating	Auxiliary circuits
DDHF	✓	✓						
DDHF/R	✓	✓						
MSND	✓	✓						
DVE			✓	✓				✓
VDA-Saturne			✓					
VCD				✓	✓			✓
DS/THF				✓		✓	✓	✓
TC				✓	✓	✓	✓	✓

- Baker Hughes can provide almost the complete set of pumps required for any plant type and size in geothermal applications.
- Reliable products are included in a wide portfolio spanning from high pressure high temperature pumps to extremely high capacity units designed to fit the process requirements.
- Best combinations of different material grades and advanced coatings are available to withstand corrosion and erosion issues depending on geothermal fluids characteristics.

Pumps for brine and condensate reinjection and ORC working fluid

Steam Turbines References for Geothermal

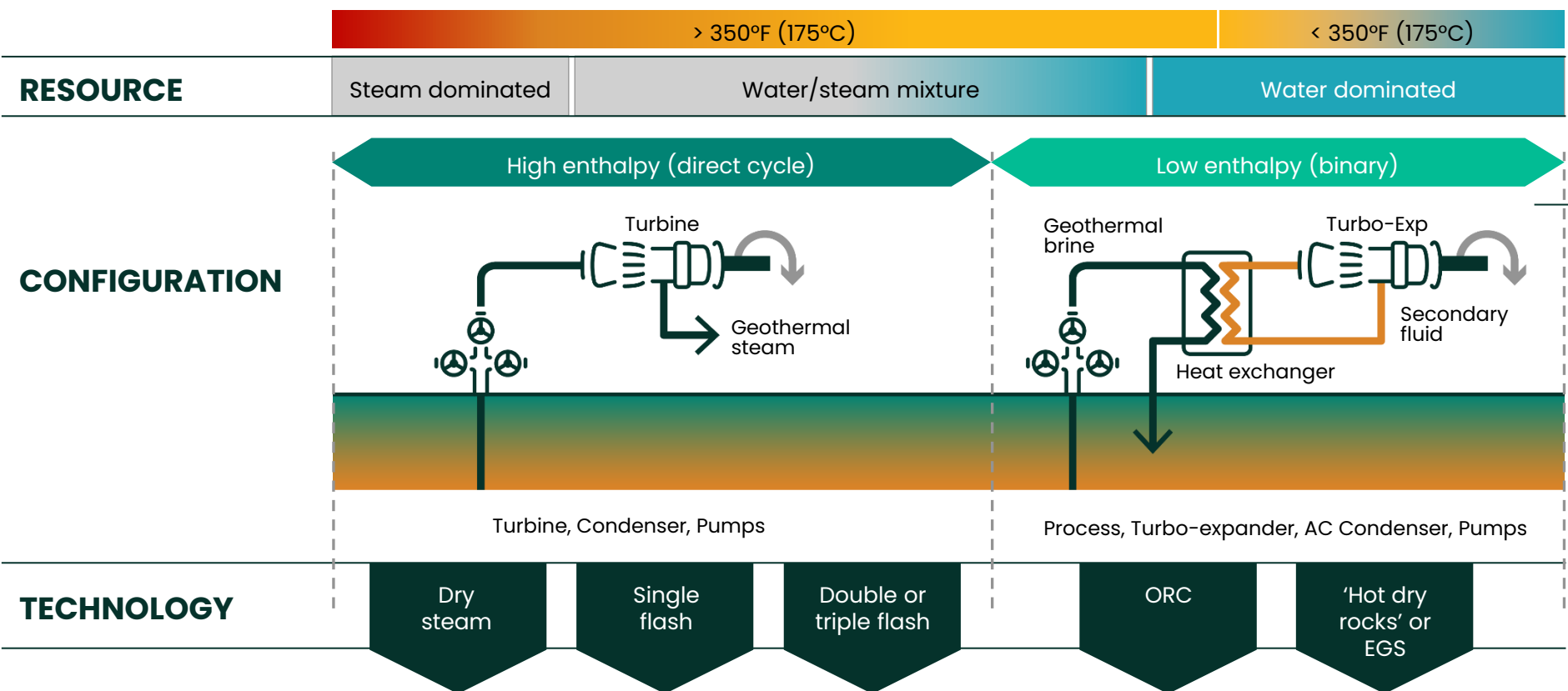
Job	Year	Model	Plant Location	Plant Type	Speed rpm	Power MW	Inlet Press. Bar-a	Inlet Temp. °C	Exhaust Press. Bar-a
1908051	2000	SGC4-23	Lago - Italy	Dry Steam	3000	10	2	120	0.061
1908053	2000	SGC4-23	Gabbro - Italy	Dry Steam	3000	20	6	159	0.075
1908054	2000	SGC4-26	Radicondoli - Italy	Single Flash	3000	40	8	200	0,065
1908056	2000	SGC4-23	Molinetto - Italy	Dry Steam	3000	20	6	190	0.065
1908057	2000	SGC4-23	Monterotondo - Italy	Dry Steam	3000	10	2	120	0.065
1908059	2001	SGC4-26	Travale - Italy	Single Flash	3000	40	8	200	0.065
1908076	2003	SGC4-26	S. Martino - Italy	Dry Steam	3000	40	8	170	0.064
1908079	2004	SGDFC3-22	Berlin - El Salvador	Single Flash	3600	44	9.5	177	0.065
1908253	2018	SGNC3-6	Imperial Valley - USA	Double Flash	3800	8.4	20	220	9

230 MW installed power...1,000,000+ operating hours

Baker Hughes 

Geothermal power generation

Thermodynamic cycle adaptation depending on resource conditions



Binary/Organic Rankine Cycle

- Operate at **lower water temperatures: 110–175 °C.**
- The **organic working fluid with a low boiling point** uses the heat from the hot water to boil.

Direct-dry steam power plants

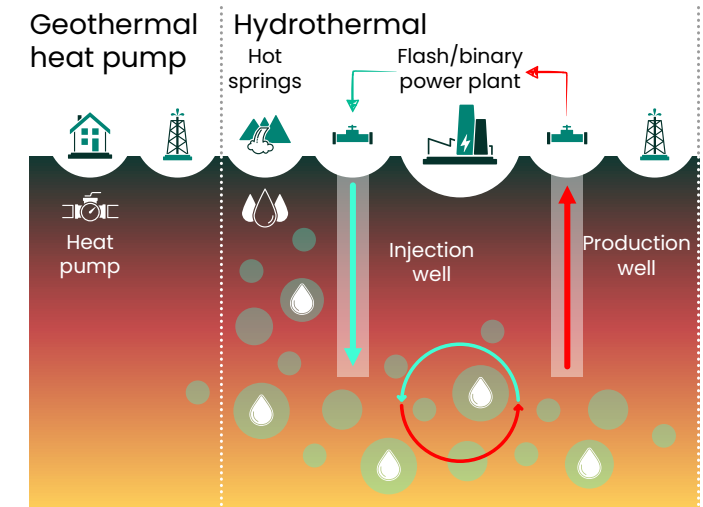
- **High enthalpy vapor-dominated resources**
- **Highest efficiency** among geothermal power plants
- **Simple** to operate
- Relatively **low capital costs**

Single / dual flash power plants

- Most common type of geothermal power plants
- **Medium- to high-enthalpy liquid-dominated resources**

BH Geothermal offering value proposition

- Complete solution from reservoir analysis to power plant
- Full integration between subsurface and surface
- Optimization of the interfaces between subsurface and surface systems
- Zero emission solutions with full NCG reinjection
- Binary cycle with zero GWP fluids
- High efficiency and reliable steam turbines
- Solutions for EGS and AGS systems
- Project financing capabilities
- Capability to execute complex projects



*NCG: Non condensable gasses

*GWP: Global Warming Potential